
**GIS based groundwater quality mapping in southeastern part of
Anantapur District, Andhra Pradesh, India**

Sunitha.V¹, Muralidhara Reddy. B², Jagadish Kumar. M³, Ramakrishna Reddy. M⁴

1- Assistant Professor, Principal Investigator – DST, Department of Geology & Geoinformatics, Yogi Vemana University, Vemanapuram, Kadapa – 516 003, A.P

2- Project Assistant – DST, Department of Geology & Geoinformatics, Yogi Vemana University, Vemanapuram, Kadapa – 516 003, A.P

3- M.Tech (Remote Sensing), H.No-7/216, Upparapalli BO, Mannur So, Balijapalli Colony Rajampet, Kadapa District-516126, A.P

4- Professor, Department of Earth Sciences, Yogi Vemana University, Vemanapuram, Kadapa – 516 003, A.P
vangala_sunitha@yahoo.com

ABSTRACT

Groundwater is used for domestic, industrial water supply and for irrigation all over the world. Although three-fourth of earth is being surrounded by sea only a little portion of it can be used for drinking purpose. The study area lies in the Southeastern part of Anantapur District, Andhra Pradesh. The area is located in the survey of India toposheets No's 57 F/14, F/15, J/3, J/4 lying between East 77°30' - 78°15' longitudes and 14°0' - 14°30' North latitudes. Geologically, it is underlain mainly by Peninsular gneisses of Archean age and comprises pink granites, schists and composite gneisses (equivalents of the Dharwars), intruded by a few pegmatite dykes. Major geomorphic units are denudational hills, dissected pediments, pediplain and alluvium. The study area experiences a semiarid climate. The study area faces water scarcity as well as water quality problems. Hence, a Geographic Information System (GIS) based assessment of spatiotemporal behaviour of groundwater quality has been carried out in the region. A surface map was prepared in the Arc GIS 9.3 (spatial analyst module) to assess the quality in terms of spatial variation and it showed that the high and low regions of water quality varied spatially during the study period. All the other parameters analysed are in adherence to desirable limits of WHO and Indian Standards for drinking water. Hence the groundwater in the study area is not suitable for drinking purpose. The spatial analysis of groundwater quality patterns of the study area shows seasonal fluctuations and these spatial patterns of physical and chemical constituents are useful in deciding water use strategies for various purposes.

Key words: Hydrogeochemistry, Spatiotemporal, GIS, Southeastern part of Anantapur District, A.P

1. Introduction

Water is the most valuable and vital resource for sustenance of life and also for any developmental activity. With the surface water sources dwindling to meet the various demands, groundwater has become the only reliable resource. The indiscriminate use of this vital natural resource is creating groundwater – mining problem in various parts of world (Todd 2007). Hence, the groundwater resource should be evaluated thoroughly, carefully and reliably on a real time basis to meet the ever growing needs. Groundwater is a natural resource for the economic and secure provision of drinking water, which plays a fundamental

role in human well being. How-ever, the pressure on groundwater, in terms of both quantity and quality, has increased to an extent whereby not only drinking water sources but also sensitive ecosystems are threatened by contamination through human exploitation. It is a well-known fact that a polluted environment has a detrimental effect on health of people, animal life and vegetation (Sujatha and Reddy 2003). Groundwater is extensively used for domestic, industrial and irrigation activities. Water quality analysis is one of the the most important issues in groundwater studies. Its monitoring and assessment is imperative for devising preventive measures against health hazards (Dhiman 2005). Quality of groundwater is equally important to its quantity owing to the suitability of water for various purposes. Variation of groundwater quality in an area is a function of physical and chemical parameters that are generally influenced by geological formations and anthropogenic activities (Subramani et al., 2005). The hydrogeochemical study reveals the zones and quality of water that are suitable for drinking, agricultural and industrial purposes (Anbazhagan and Archana 2004). The chemical quality of groundwater is related to the lithology of the area and the residence time of the water in contact with rock material. Weathered mantle, soils and atmosphere are the important factors responsible for contribution of dissolved solids to water. It is impossible to control the dissolution of undesirable constituents in the waters after they enter the ground (Gowd 2005).

Geographical Information System (GIS) based groundwater studies were concentrated on the preparation of hydrogeomorphological maps, interpretation of lineaments and integrated terrain analysis (Kamaraju 1997; Srivastava and Bhattacharya 2000; Khan and Moharana 2002). Now GIS is increasingly being integrated with groundwater and surface water quality assessment models. GIS can be used as tool for spatial surveillance of epidemiological diseases and also as a spatial decision support system for public health management. The spatial patterns of chemical constituents are useful in deciding the water use strategies for various purposes (Collins et al 1998; Gogu et al., 2001; Dhiman 2005; Shankar et al., 2010). Assessing and monitoring the quality of groundwater is therefore, important to ensure sustainable safe use of these resources for the various purposes. In the study area, the data pertaining to the groundwater quality are not recorded up to till the time of this study. Hence the present study is an attempt to understand and identify the spatiotemporal behavior of groundwater quality of southeastern part of Anantapur District, Andhra Pradesh.

2. Study Area

The study area (Figure 1) in the southeastern part of Anantapur District of Andhra Pradesh is in between East $77^{\circ}30'$ - $78^{\circ}15'$ longitudes and $14^{\circ}0'$ - $14^{\circ}30'$ North latitudes and falls in the Survey of India Toposheet Nos. 57 F/14, F/15, J/3, J/4. The major geomorphic units are denudational hills, dissected pedimonts, pediplain and alluvium. Geologically, it is underlain mainly by Peninsular gneisses of Archean age and comprises pink granites, schists and composite gneisses (equivalents of the Dharwars), intruded by a few pegmatite dykes, numerous dolerite dykes and volcanic pipes. It experiences a semiarid climate, with a moisture index of 33.7 % and monthly mean temperatures a low of 15°C in January and high of 39°C in May. The normal rainfall of the district is 553 mm that is the least, when compared to that in the other districts of Andhra Pradesh.

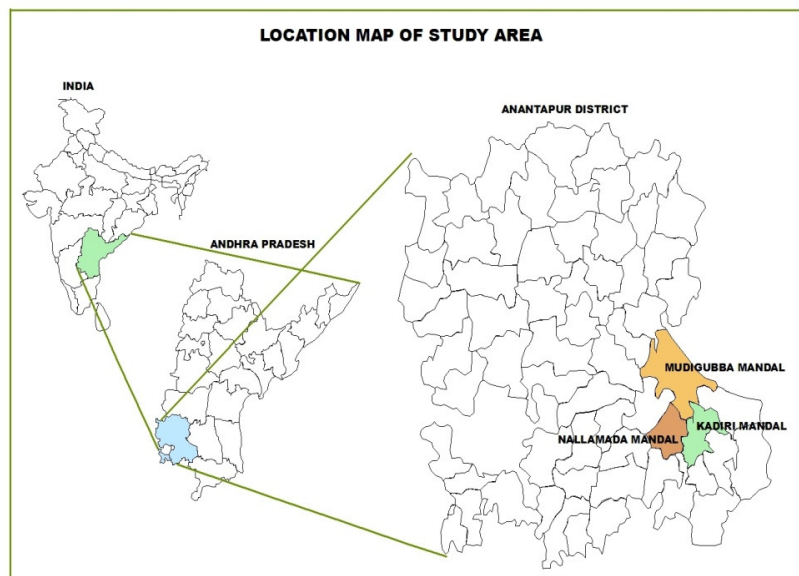


Figure 1: Study area location map

3. Materials and methods

In the present study, water samples were collected from fifty tube wells in the study area. The study is carried out with the help of topographic sheets, Garmin GPS MAP 76, Arc GIS 9.3 and fieldwork. Toposheets are used to prepare the basemap, drainage map and to understand the general nature of the study area. GPS is used to map the location of each sampling well and finally the results were taken to the GIS for further analysis. The field work included water level measurements, well inventory and collection of water samples from tube wells and the study of geological and geomorphological features of the area in general. The pre-monsoon water samples were collected during September 2009 and the post monsoon water samples were collected during April 2010. The groundwater samples were analyzed, as per the procedure of APHA (1995), and suggested precautions were taken to avoid contamination. The various parameters determined were: pH, total hardness (TH), chloride (Cl^-), nitrate (NO_3^-) and fluoride (F^-). pH was determined by pH meter; TH, Cl^- by titrimetry; F^- and NO_3^- were determined by using ion selective electrode (Orion 4 star ion meter, Model: pH/ISE). The location of each wells were taken into the GIS environment and the results of each parameters analysed were added to the concerned wells. Spatial analyst, an extended module of Arc GIS 9.3 was used to find out the spatiotemporal behavior of the groundwater quality parameters.

4. Results and Discussion:

4.1 Chemical Analysis

The results of chemical and statistical parameters analyzed are given in Table 1. The statistical parameters include minimum, maximum, mean and standard deviation. Understanding the quality of groundwater is as important because it is the main factor determining its suitability for drinking, domestic, agricultural and industrial purposes. The pH values of groundwater range from 7 to 8.6 with an average of 7.98 during September 2009. However, during April 2010 it ranges from 7 to 8.6 with an average value of 7.84. This shows that the groundwater of the study area is mainly alkaline in nature. The total hardness values range from 246 to 1474 mg/l during September 2009 and 413 to 1898 mg/l during

April 2010. Chlorides ranged between 72 and 480 mg/l for September 2009 and 70 to 460 mg/l during April 2010, respectively. The fluoride values ranged from 1.1 to 7.1 mg/l during September 2009 and April 2010. Nitrate (NO_3^-) in the groundwater varies from 20 to 110 mg/l during September 2009 and 20 to 100 mg/l during April 2010.

Table 1: Result of chemical analysis and statistical parameters of groundwater samples collected from the study area

S.N o	P^{H}		TH (mg/l)		Cl^- (mg/l)		F^- (mg/l)		NO_3^- (mg/l)	
	Sep 09	April 10	Sep 09	April 10	Sep 09	April 10	Sep 09	April 10	Sep 09	April 10
W ₁	7.4	7.2	678	712	140	130	3.5	3.4	58	54
W ₂	7.6	7.4	650	696	170	170	4.2	4.0	55	50
W ₃	7.3	7.2	655	705	190	180	4.0	3.8	60	58
W ₄	7.5	7.2	346	413	170	160	5.0	4.8	68	65
W ₅	8.3	8.0	413	471	80	75	5.2	5.0	90	85
W ₆	8.2	8.2	496	529	82	80	1.4	1.2	92	92
W ₇	8.3	8.0	246	1030	190	180	1.1	1.0	70	70
W ₈	8.3	8.3	505	530	95	90	4.3	4.2	30	28
W ₉	8.3	8.0	538	555	98	95	3.8	3.6	62	60
W ₁₀	8.4	8.2	596	637	110	100	3.3	3.0	45	40
W ₁₁	7.2	7.0	662	695	120	110	4.2	4.0	48	45
W ₁₂	7.0	7.0	728	787	130	120	4.6	4.4	56	50
W ₁₃	8.0	7.6	630	696	134	128	3.3	3.0	70	68
W ₁₄	8.3	8.3	621	754	220	130	3.5	3.2	72	70
W ₁₅	8.4	8.4	637	736	210	210	3.8	3.6	56	54
W ₁₆	8.5	8.5	494	703	190	200	2.2	2.0	48	40
W ₁₇	8.6	8.4	478	536	94	190	1.9	2.0	98	95
W ₁₈	7.9	7.2	490	462	96	90	1.6	1.6	78	70
W ₁₉	8.2	8.0	596	471	110	90	2.2	2.0	65	65
W ₂₀	8.3	8.3	530	580	124	100	1.8	1.8	60	60
W ₂₁	8.0	8.2	539	534	128	120	3.4	3.2	20	78
W ₂₂	8.6	8.6	580	564	330	110	3.6	3.4	70	70
W ₂₃	7.5	7.2	639	622	410	320	3.0	3.0	100	95
W ₂₄	7.6	7.5	540	623	210	400	2.8	2.8	46	45
W ₂₅	8.2	7.6	398	456	180	150	2.4	2.2	56	50
W ₂₆	8.6	8.2	480	514	170	160	2.8	2.6	90	85
W ₂₇	8.0	8.4	539	584	130	120	2.1	2.0	97	90
W ₂₈	7.6	8.0	621	655	170	150	3.5	3.4	88	85
W ₂₉	8.0	7.2	605	638	178	170	2.2	2.0	70	70
W ₃₀	8.2	7.8	704	696	180	140	2.8	2.6	66	65
W ₃₁	8.3	8.0	721	741	210	200	7.1	7.1	20	20
W ₃₂	8.0	8.0	537	562	220	210	4.4	4.2	68	65
W ₃₃	7.4	7.2	562	595	240	230	2.7	2.6	49	45
W ₃₄	7.6	7.4	528	557	160	150	2.6	2.5	56	50
W ₃₅	7.2	7.6	478	503	72	70	3.4	3.2	70	68
W ₃₆	8.0	7.8	548	561	240	220	4.3	4.3	77	70
W ₃₇	8.2	8.0	594	619	260	250	4.6	4.5	56	50
W ₃₈	8.6	8.2	628	665	270	260	5.2	5.0	72	70
W ₃₉	8.3	8.6	648	694	290	270	4.6	4.4	54	52
W ₄₀	8.0	8.3	744	827	360	350	4.3	4.3	80	75
W ₄₁	8.2	7.8	835	902	310	300	3.6	3.6	80	75
W ₄₂	7.6	8.0	910	1000	280	270	3.8	3.8	63	60

W ₄₃	7.8	7.6	926	1017	210	200	3.7	3.5	33	30
W ₄₄	7.6	7.8	969	964	220	200	4.5	4.4	35	35
W ₄₅	8.0	7.6	1067	1133	320	300	3.7	3.7	32	30
W ₄₆	8.2	8.0	1124	1219	410	400	3.8	3.8	90	85
W ₄₇	8.3	8.2	1240	1306	378	370	4.0	3.8	100	90
W ₄₈	8.5	8.3	1322	1413	480	460	4.2	4.0	110	95
W ₄₉	8.0	8.0	1333	1898	450	290	4.5	4.2	90	100
W ₅₀	8.2	7.8	1474	1479	380	300	4.0	4.5	78	90
Min	7.0	7.0	246	413	72	70	1.1	1	20	20
Max	8.6	8.6	1474	1898	480	460	7.1	7.1	110	100
Mean	8.0	7.86	676.4	744.7	211.9	195.3	3.53	3.4	65.94	64.14
S.D	0.40	0.44	256.8	295.6	102.8	94.8	1.12	1.12	21.02	19.75

The analytical results of physical and chemical parameters of groundwater were compared with the standard guideline values as prescribed by the World Health Organization (WHO 1983) and Indian Standards (ISI 1983) for drinking and public health purposes (Table 2). The table shows the most desirable limits of various parameters and it was found that all the parameters analyzed are above the desirable limits of both standards.

Table 2: A comparison of water quality parameters (drinking) of the study area with WHO and Indian Standards

Parameters	WHO International Standard (1983)	Indian Standards for drinking water specification	Mean value of parameters under analysis	
			Sep 09	April 10
pH	7-8.5	6.5-8.5	8.0	7.8
TH (mg/l)	100	300	676.4	744.7
Cl ⁻ (mg/l)	200	250	211.9	195.3
F ⁻ (mg/l)	1.5	1.5	3.53	3.40
NO ₃ ⁻ (mg/l)	45	45	65.94	64.14

4.2 Spatial analysis of groundwater quality

The GIS based analysis of spatiotemporal behavior of the groundwater quality in the study area was done using the spatial analyst module of Arc GIS 9.3 powerful spatial analysis is feasible once the database is established. The interpolation technique used in the analysis is inverse distance weighted (IDW) method. IDW is an algorithm for spatially interpolating or estimating values between measurements. Each value estimated in an IDW interpolation is a weighted average of the surrounding sample points. Weights are computed by taking the inverse of the distance from an observations location to the location of the point being estimated (Burrough and MC Donnell 1998). The inverse distance can be raised to a power (e.g. linear, squared, and cubed) to model different geometries (e.g. line, area, volume) (Guan et al., 1999). In a comparison of several different deterministic interpolation procedures, Burrough and Mc Donnell (1998) and Mathes et al., (2006) found that using IDW with a squared distance term yielded results most consistent with original input data. This method is suitable for datasets where the maximum and minimum values in the interpolated surface commonly occur at sample points (ESRI 2002).

The spatial pattern of parameters was analysed in two seasons, i.e. pre monsoon (April 2010) and post monsoon seasons (September 2009). The spatial analysis of pH during the two seasons shows the change in the areas of spatial highs and lows (Figure 2). During September 2009, the spatial highs are showing a clear NW-SE alignment, but in April 2010 it shows an eastward shift with a spatial maximum in the northwestern side. The total hardness highs are more prominent in September 2009 while compared to April 2010. The abundance of spatial highs during September 2009 points towards indicate high rate of rainfall. This hardness is attributed to the use of fertilizers and pesticides in the plantation (Figure 3). The spatial distributions of chlorides (Cl^-) highs are very less in September 2009 but in April 2010 the distribution of chloride highs are prominent (Figure 4). However, in the case of NO_3^- and F^- , the spatial highs show a spread in September 2009, and during April 2010, the isolated patches of highs are distributed in the area (Figure 5&6).

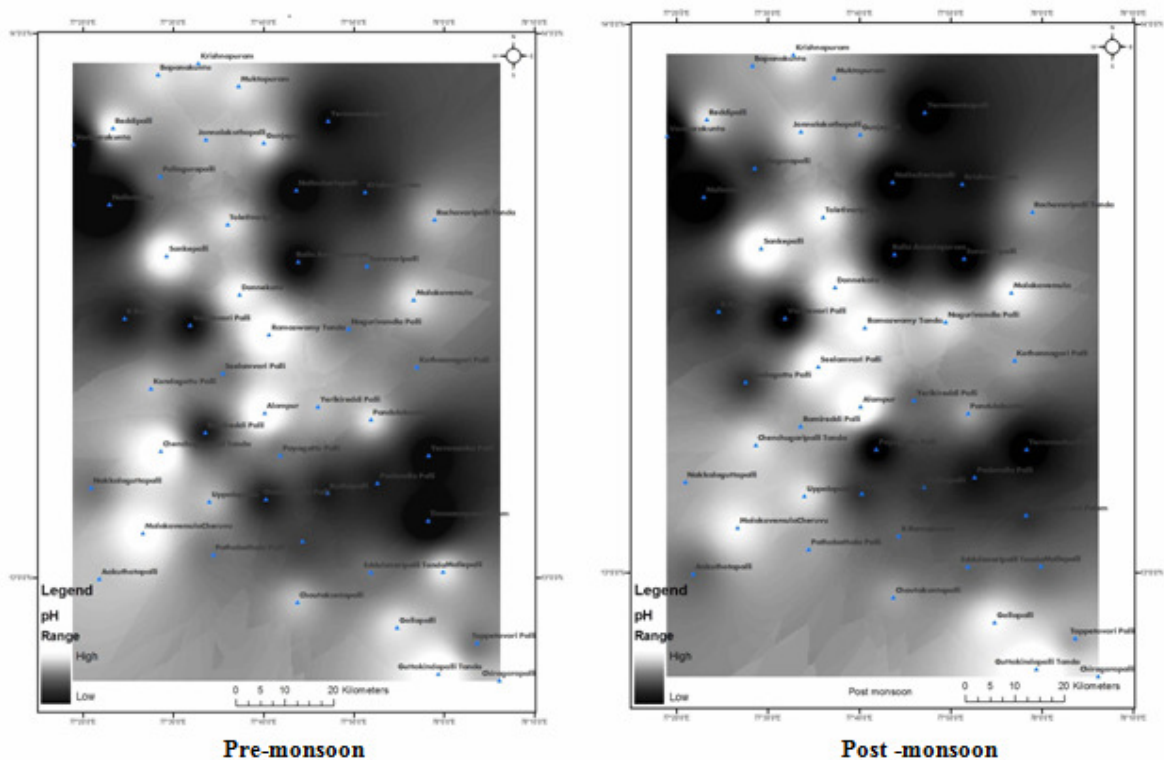


Figure 2: Maps showing spatiotemporal variation of pH in the study area

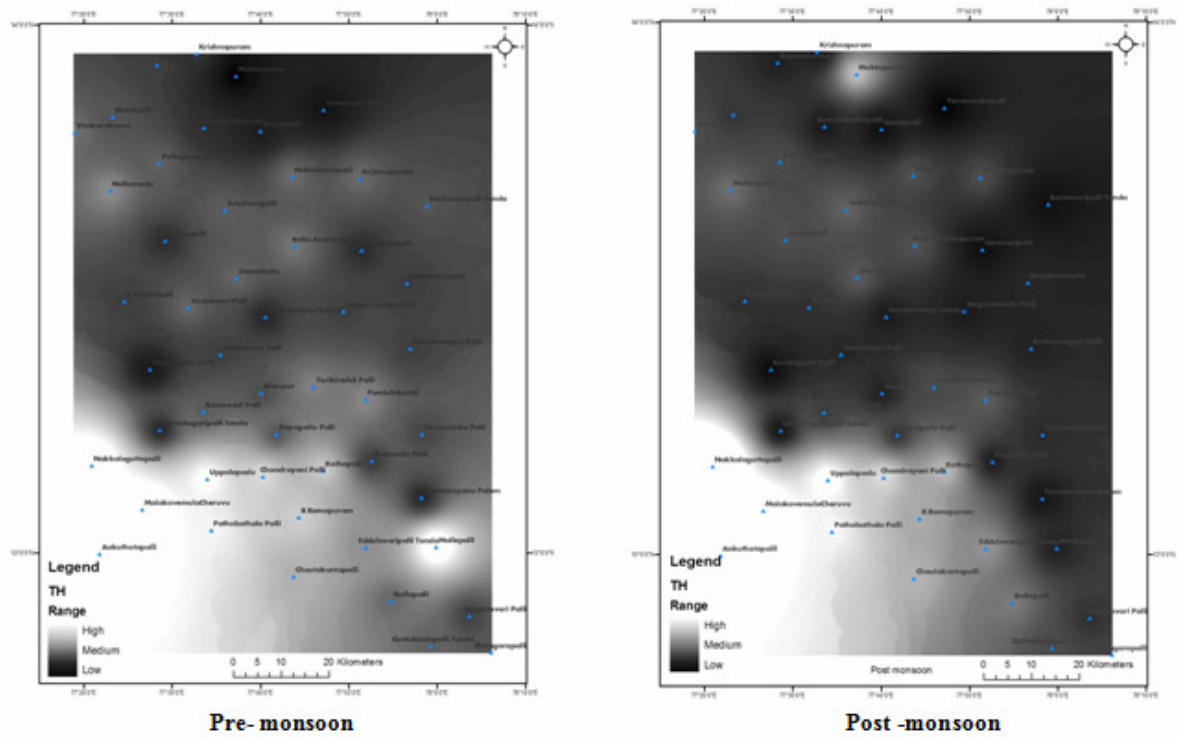


Figure 3: Maps showing spatiotemporal variation of TH in the study area

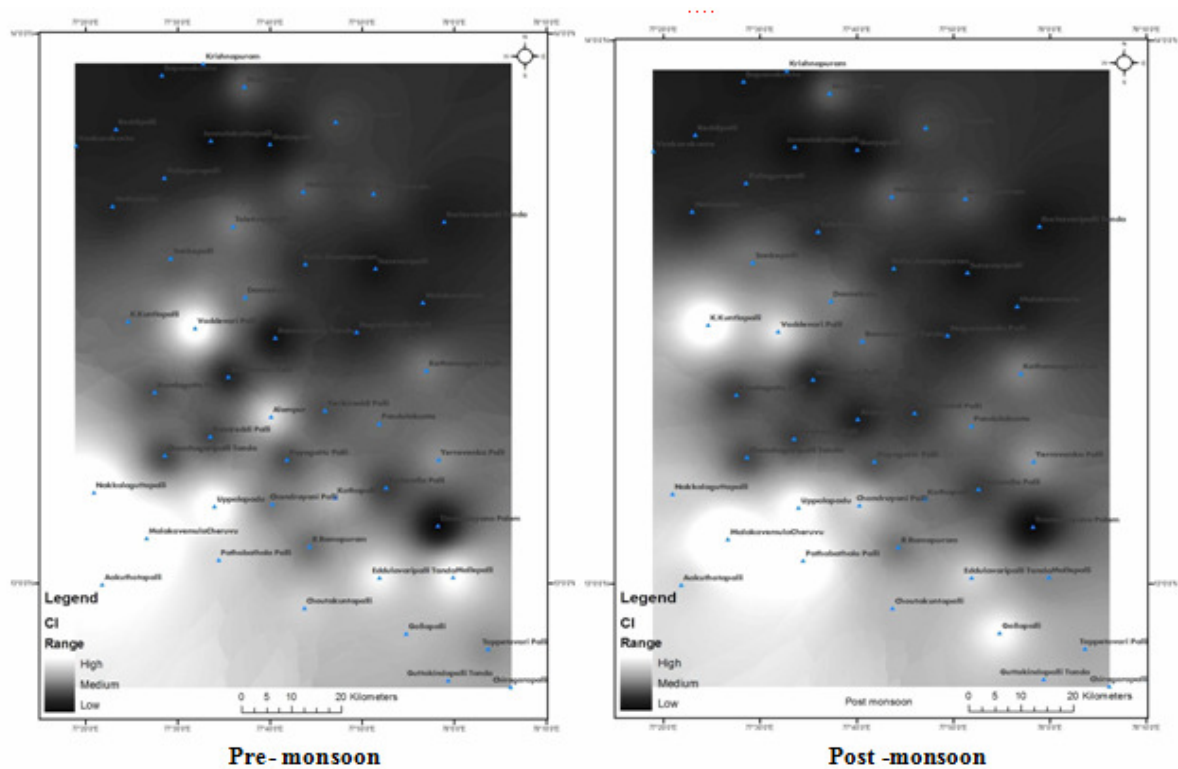


Figure 4: Maps showing spatiotemporal variation of Cl⁻ in the study area

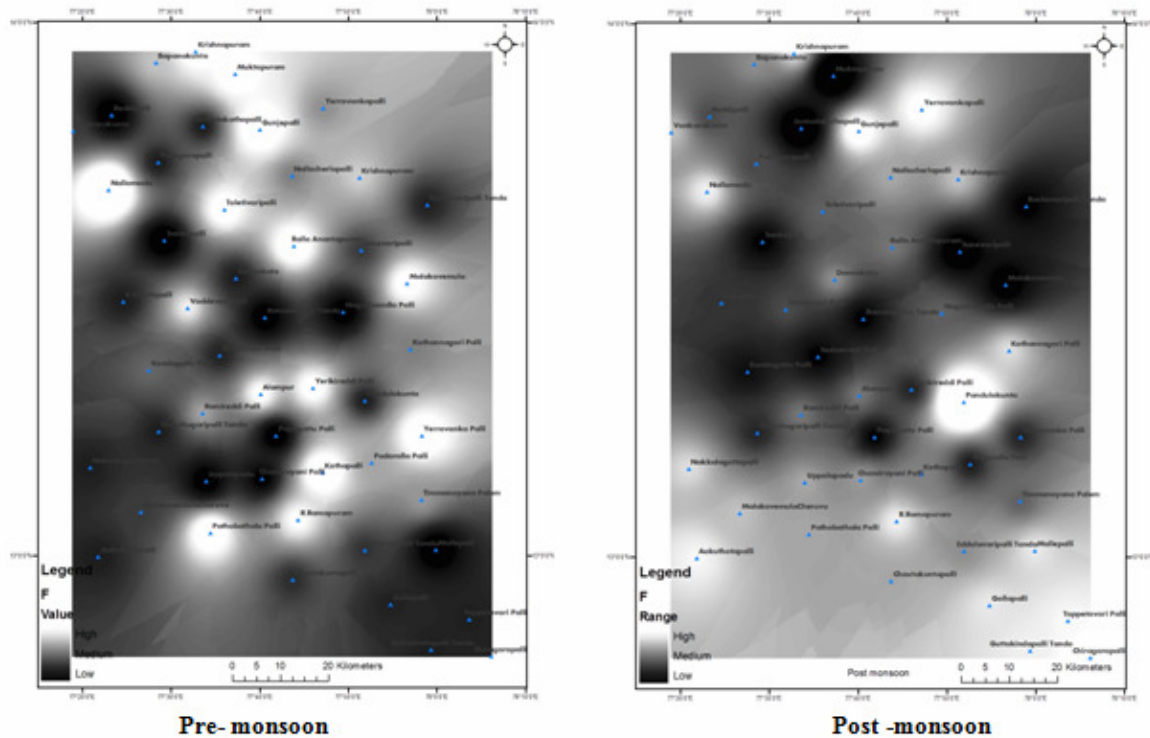


Figure 5: Maps showing spatiotemporal variation of F^- in the study area

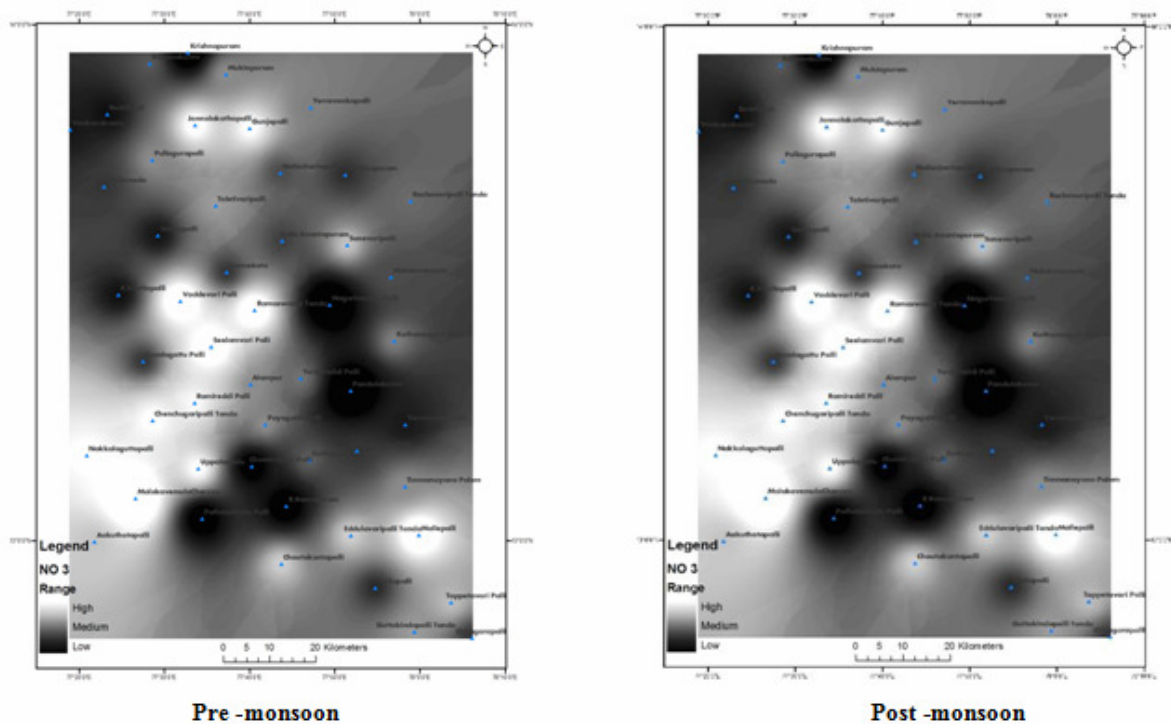


Figure 6: Maps showing spatiotemporal variation of NO_3^- in the study area

5. Conclusion

The purpose of present study area was to understand the groundwater quality in southeastern part of Anantapur district, Andhra Pradesh and pictorially represent it using GIS. The

chemical characteristics of the water samples reveal slight seasonal variations. The waters of the region are alkaline. All the other parameters analysed are above the desirable limits of WHO and Indian Standards for drinking water. Hence the groundwater in the study area is not suitable for drinking purpose. The post monsoon groundwater samples show maximum deviation, as this is the time when precipitation is less so that the concentration of ions will increase considerably. The influence of lithology over the quality of groundwater is ruled out because majority of the area comes under single lithology i.e. granitic gneiss and it does not have much influence over the quality of water. The spatial highs of pH are showing a clear alignment in pre-monsoon season but in post-monsoon season it shows an eastward shift with a spatial maximum in the northwestern side. The spatial distributions of chloride (Cl^-) highs are very prominent in post-monsoon season when compared to pre-monsoon season. However, in the case of NO_3^- and F^- , the spatial highs show a spread in post monsoon season and during pre-monsoon season the isolated patches of highs are distributed in the area. An excessive use of fertilizers and pesticides in agriculture practice was noticed during the field work. Hence proper attention should be given to water quality monitoring along with besides groundwater resource development.

Acknowledgments

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